MATH3871/5960 2017 mid-session solutions

Question 1 (11 Marks)

Consider the problem of expressing subjective prior beliefs about parameters values via prior distributions in Bayesian statistical inference. Give **brief** descriptions in words for the following questions:

a) **[3 marks]** What is a conjugate prior? Why do we use conjugate priors and when is it not appropriate to use them?

- If a prior belongs to a family of distributions P such that the posterior also belongs to P, then it is a conjugate prior.
- Usage: to help obtain closed form posteriors
- Do not use: When the prior family is not flexible enough to describe your prior beliefs.

b) **[3 marks]** What is an improper prior? Give an example of one and state the conditions under which an improper prior can be used for inference.

- An improper prior $\pi(\theta)$ is such that $\int \pi(\theta)d\theta = \infty$.
- E.g. for a location parameter we might have $\pi(\theta) \propto 1$ on $\mathbb{R}$.
- We can use the prior for (single model) inference if the resulting posterior is proper.

c) **[2 marks]** What is Jeffreys' prior and when is it used?

- Jeffreys' prior is defined as $\pi_J(\theta) = \sqrt{|I_n(\theta)|}$, the square root of determinant of the information matrix.
- It is used when one wishes to express weak prior information that is equivalent across parameter transformations.

d) **[3 marks]** Consider a (general) mixture prior distribution for θ of the form

$$\pi(\theta) = \sum_{i=1}^{k} w_i \pi_i(\theta) \qquad \sum_{i=1}^{k} w_i = 1.$$

Give an example of when the mixture prior distribution may become an appropriate expression for the prior belief. What is the role of the weights w_i? Do the distributions $\pi_i(\theta)$ have to take the same form?

- E.g. when spinning a coin. Some bias towards heads or tails is expected, but exactly which of these is not known. Hence we need a mixture prior to cover both possibilities.
- The weights define the relative contributions to the overall prior given by each component.
- No, they can be completely different.

Question 2 (20 Marks)

Consider a vector $\mathbf{y} = \{y_1, \ldots, y_n\}$ of n iid observations from the distribution $N(\mu, \sigma^2)$, where mean μ and variance σ^2 are unknown. Then each y_i has probability density function of the form

$$f(y_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(y_i - \mu)^2)$$

Suppose we take the vague prior density for μ and σ given by

$$\pi(\mu, \sigma^2) \propto (\sigma^2)^{-1}$$

and let

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

a) **[5 marks]** Derive the joint posterior density $\pi(\mu, \sigma^2|\mathbf{y})$, show that it is proportional to

$$\sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y} - \mu)^2]\right).$$

$$\begin{aligned}
\pi(\mu, \sigma^2|y) &\propto \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2\right) \\
&= \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}\left[\sum_{i=1}^{n}(y_i - \bar{y})^2 + n(\bar{y} - \mu)^2\right]\right) \\
&= \sigma^{-n-2} \exp(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y} - \mu)^2])
\end{aligned}$$

where

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

is the sample variance of the y_i's.

b) **[3 marks]** Show that the conditional distribution $\pi(\mu|\sigma^2, \mathbf{y})$ is of the form

$$\mu|\sigma^2, \mathbf{y} \sim N(\bar{y}, \sigma^2/n).$$

$$\begin{aligned}
\pi(\mu|\sigma^2, y) &\propto \sigma^{-n-2}\exp(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y}-\mu)^2]) \\
&\propto \exp(-\frac{1}{2\sigma^2}n(\bar{y}-\mu)^2) \\
\mu|\sigma^2, y &\sim N(\bar{y}, \sigma^2/n).
\end{aligned}$$

c) **[5 marks]** Show that the marginal posterior distribution σ^2 is

$$\pi(\sigma^2|\mathbf{y}) \propto (\sigma^2)^{-(n+1)/2}\exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right).$$

$$\begin{aligned}
\pi(\sigma^2|y) &\propto \int \sigma^{-n-2}\exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y}-\mu)^2]\right)d\mu \\
&\propto \sigma^{-n-2}\exp\left(-\frac{1}{2\sigma^2}(n-1)s^2\right)\int \exp\left(-\frac{1}{2\sigma^2}n(\bar{y}-\mu)^2\right)d\mu \\
&\propto \sigma^{-n-2}\exp\left(-\frac{1}{2\sigma^2}(n-1)s^2\right)\sqrt{2\pi\sigma^2/n} \\
&\propto (\sigma^2)^{-(n+1)/2}\exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)
\end{aligned}$$

d) **[7 marks]** Show that the marginal posterior distribution for μ is

$$\pi(\mu|\mathbf{y}) \propto \left[1 + \frac{n(\bar{y}-\mu)^2}{(n-1)s^2}\right]^{-n/2}.$$

Hint: You might like to make the substitutions

$$z = \frac{A}{2\sigma^2}, \quad A = (n-1)s^2 + n(\bar{y}-\mu)^2$$

and observe that the normalised Gamma distribution has the form

$$f(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\theta^{\alpha-1}e^{-\beta\theta}, \theta > 0.$$

$$\pi(\mu|y) \propto \int \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y}-\mu)^2]\right) d\sigma^2$$

Substitute $z = A/(2\sigma^2)$ where $A = (n-1)s^2 + n(\bar{y}-\mu)^2$.
$dz = -\frac{A}{2\sigma^4}d\sigma^2$, $\sigma^2 = A/(2z)$
Hence

$$\begin{aligned}
\pi(\mu|y) &\propto \int \sigma^{-n-2} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y}-\mu)^2]\right) d\sigma^2 \\
&= \int (A/2z)^{-(n+2)/2} \exp(-z) \frac{2(A^2/(2z)^2)}{A} dz \\
&\propto A^{-n/2} \int z^{(n-2)/2} \exp(-z)\, dz \\
&\propto \left((n-1)s^2 + n(\bar{y}-\mu)^2\right)^{-n/2} \\
&\propto \left(1 + \frac{n(\bar{y}-\mu)^2}{(n-1)s^2}\right)^{-n/2}
\end{aligned}$$

Question 3 (10 Marks)

In Lectures, you were given the rejection algorithm, which obtains a sample x from the distribution $f(x)$ via the following three steps:

1 Simulate x from $g(x)$

2 Simulate y from $U(0, Kg(x))$

3 Accept x if $y \leq f(x)$

a) **[2 marks]** The function $g(x)$ can be chosen arbitrarily, however in practice, a good choices of g is essential to the performance of rejection sampling. Give two features of $g(x)$ that are either essential or desirable for the algorithm.

- g must have a support that covers the support of f (i.e. $g(\theta) > 0$ if $f(\theta) > 0$.)
- The acceptance probabilities are higher if g is closer to f
- Must be able to find a value $K > 0$ such that $Kg(\theta) \geq f(\theta)$ for all θ

b) **[2 marks]** Why is K needed in the algorithm?

For the algorithm to work we need to have $g(\theta) \geq f(\theta)$ for all θ. Clearly this won't be the case except for $g(\theta) = f(\theta)$. So we introduce a constant K so that $Kg(\theta) \geq f(\theta)$ is easily achievable.

c) **[4 marks]** What is the probability of accepting a single point x in the rejection step? Hence derive the overall acceptance probability for all samples simulated from $g(x)$.

If $\theta^* \sim g(\theta)$, then the probability of accepting θ^* is $h(\theta^*) = f(\theta^*)/Kg(\theta^*)$.

Also

$$Pr(\text{accept any sample}) = \int h(\theta)g(\theta)d\theta = \frac{1}{K}\int f(\theta)d\theta = \frac{1}{K}.$$

d) **[2 marks]** Hence, using the acceptance probability given in part d), give an alternative algorithm (to the above) for rejection sampling.

- Sample $\theta^* \sim g(\theta)$
- Accept θ^* as a draw from $f(\theta)$ with probability $f(\theta^*)/Kg(\theta^*)$.

Question 4 (14 Marks)

Consider the loss function $L(\theta, d)$ which quantifies the loss achieved when making decision $d \in \mathcal{D}$ given a fixed parameter value θ.

1. **[2 marks]** What is the role of the loss function in Bayesian inference? Write down an expression for the optimal decision, d^*, in this setting.

   - The role is to assign a numerical value to the benefits or loss in making a decision d based on the posterior distribution. This then allows an optimum decision to be made.
   - $d^* = \arg\min_d E_{\pi(\theta|x)}[L(\theta, d)]$ i.e. the decision that leads to the smallest posterior expected loss.

2. **[6 marks]** Show that adopting the quadratic loss $L(\theta, d) = (\theta - d)^2$ results in the optimal parameter estimate $d^* = E(\theta|x)$, the posterior mean.

$$\begin{aligned} E_\pi[L(\theta, d)] &= \int L(\theta, d)\pi(\theta|x)d\theta \\ &= \int (\theta - d)^2\pi(\theta|x)d\theta \\ &= \int (\theta - E(\theta|x) + E(\theta|x) - d)^2\pi(\theta|x)d\theta \\ &= \int (\theta - E(\theta|x))^2\pi(\theta|x)d\theta + \int (E(\theta|x) - d)^2\pi(\theta|x)d\theta \\ &\qquad +2\int (\theta - E(\theta|x))(E(\theta|x) - d)\pi(\theta|x)d\theta \\ &= \mathrm{Var}(\theta|x) + (E(\theta|x) - d)^2 + 0 \end{aligned}$$

   This is minimised when $d = E(\theta|x)$ is the posterior mean.

3. **[6 marks]** Derive the optimal parameter estimate, d^*, when using the 0-1 loss function

$$L(\theta, d) = \begin{cases} 0 & \text{if } |\theta - d| \leq \epsilon \\ 1 & \text{if } |\theta - d| > \epsilon \end{cases}.$$

   Here

$$\begin{aligned} E[L(\theta, d)] &= 0 \times \Pr(|\theta - d| \leq \epsilon) + 1 \times \Pr(|\theta - d| > \epsilon) \\ &= 1 - \Pr(|\theta - d| \leq \epsilon). \end{aligned}$$

- $|\theta - d| \leq \epsilon$ defines an interval $[\theta - \epsilon, \theta + \epsilon]$ of length 2ϵ
- To minimise, choose interval with highest probability i.e. high density region
- Then θ is mid-point of interval with highest probability
- Choosing ϵ arbitrarily small will select the posterior mode for d.

$\Rightarrow$ The posterior mode minimises 0-1 loss.